

Nancy Bhardwaj

AI Quality Engineering Lead | Test Automation Architect | Agentic AI QA Engineer

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PROFESSIONAL SUMMARY

AI Quality Engineering Lead and Test Automation Architect with **14+** years of experience building enterprise test automation, API quality pipelines, and AI-powered QA solutions. Led distributed teams of **up to 15 engineers** and architected **Playwright (TypeScript)** frameworks that reduced regression execution time by **70%**. Hands-on experience in **Salesforce Agentforce testing, LLM and RAG evaluation**, AI-agent behavior validation, AI red teaming and human-in-the-loop review design. Built practical **AI QA platforms and agentic workflows** using **Python, FastAPI, n8n, LangFlow, MCP, vector databases, RAGAS, Promptfoo, and DeepEval**.

TECHNICAL SKILLS

AI-Augmented QA & Agentic Automation: AI Agent Design & Orchestration | Multi-Agent Systems | n8n | LangFlow | Prompt Engineering | LLM Integration (Claude, Groq, OpenAI) | Advanced RAG Pipeline | Vector Databases (ChromaDB, Pinecone) | Human-in-the-Loop Design

GenAI/LLM Evaluation & Responsible AI Testing: RAGAS (Faithfulness, Groundedness, Context Relevance/Precision/Recall, Answer Relevance) | Promptfoo | DeepEval

Test Automation Frameworks: Playwright | Selenium WebDriver

Programming & Scripting: TypeScript | Python

API Testing & Services: REST API Testing | Postman | API Mocking

Testing Expertise: Functional | Regression | Exploratory | Mobile App | Responsive | Risk-Based Testing

Salesforce Testing: Sales Cloud | Service Cloud | Agentforce Testing (Topics, Actions, Utterances, Testing Center, Subagents) | CRM & CPQ Testing | Case Management

CI/CD & DevOps: GitHub Actions | Jenkins | Azure DevOps | Power Automate

Performance Testing: JMeter

Reporting & Analytics: Allure | Playwright Trace Viewer | Newman

Test Management: JIRA | Azure Test Plans

Databases: MySQL

Version Control: Git | GitHub

Methodologies: Agile/Scrum | SAgE | Shift-Left Testing

AI & Agentic AI Projects

QAVentra - Self-Hosted Hybrid RAG Knowledge Platform for QA Teams

Technologies: Python, FastAPI, BAAI BGE-M3 (Hybrid Embeddings), Qdrant, BGE-Reranker-v2-M3, Groq, Docker Compose, JIRA REST API, LangChain Text Splitters, MCP (Model Context Protocol), APScheduler, DeepEval

- Built a self-hosted **RAG knowledge platform** that lets QA teams search test cases, automation code, JIRA tickets, requirements, and meeting notes from a single cited knowledge base.
- Implemented **hybrid search and reranking** to improve the relevance of results for QA and defect-related queries.
- Automated knowledge-base updates with **scheduled ingestion and change detection**, keeping content current without manual updates.
- Packaged the platform using **Docker Compose** and exposed it through **MCP**, allowing AI tools and a web chat interface to query the QA knowledge base.
- Evaluated RAG response quality using **DeepEval** for relevance, faithfulness, contextual precision/recall, and citation accuracy.

SprintReadyAI - QA Refinement Copilot

Technologies: n8n, OpenAI (LLM), Slack, Human-in-the-Loop Design, JIRA, Promptfoo

- Built an **AI-powered QA refinement agent** in n8n that fetches **JIRA story details** and analyzes them from a QA perspective.
- Generated **readiness scores, missing acceptance criteria, QA clarification questions, test scenarios, regression impact, and QA estimates** to support sprint refinement.
- Integrated **Slack** to share AI-generated refinement reports with the team for **human QA review** before adding final comments to JIRA.
- Added **Promptfoo evaluation** to validate agent responses across **well-defined, vague, invalid, and missing story inputs**, improving response reliability and consistency.

TestCase Compass - Advanced RAG Pipeline

Technologies: LangFlow, Groq (LLM), Mistral Embeddings, ChromaDB, NVIDIA Rerank, HyDE, Contextual Compression, Parent-Document Retrieval, RAGAS

- Built an **Advanced RAG assistant** over **5,000+ test cases** using HyDE, Mistral embeddings, ChromaDB, NVIDIA reranking, and parent-document retrieval.
- Improved **retrieval relevance and response accuracy** using contextual compression and grounded prompts.
- Implemented **RAGAS evaluation** to measure **Faithfulness, Answer Relevancy, Context Precision, Context Recall, and Answer Correctness**.

AssertPilot - AI Agent QA and Security Observatory

Technologies: React, Vite, Python, FastAPI, MCP

- Built a working **AI Agent QA & Security Observatory prototype** to validate **agent behavior beyond the final response**, including **tool selection, parameters, execution paths, policy compliance, and risk**.
- Designed an **expected-vs-actual behavior validation framework** with security scenarios covering **prompt injection, unauthorized data access, privilege escalation**, and high-impact actions such as **unauthorized refunds**.
- Developed a **policy-driven risk and findings model** to classify agent behavior, identify **security and business-rule violations**, and provide actionable recommendations for **safer AI-agent execution**.

EMPLOYMENT

Team Lead

August 2021 to Present

QualityAI (Qualitest), Noida

Projects: Baxter Credit Union, Imperial Brands, Genesys, Keyloop

- Led projects adhering to Agile methodologies, actively participating in **Daily Stand-ups, Grooming, Planning, Retrospective, and Demo meetings**.
- Led distributed QA teams of up to 15 engineers, enhancing team productivity and performance.
- Successfully led and managed distributed teams across multiple time zones, ensuring seamless communication, collaboration, and project delivery.
- Designed and executed utterance-based test suites in **Salesforce Agentforce Testing Center**, authoring positive and negative utterances per topic to validate intent classification, action-sequence execution, and response accuracy using CSV-based batch testing and Topic/Action/Response pass-rate analysis to catch misrouting before production.
- Led the initiative to set up **Newman reporting** for enhanced visibility and analysis of API testing results.
- Led the initiative to set up **CI/CD pipelines in Azure DevOps**, integrating Newman to automate API test execution and reporting.
- Automated Email/Teams notifications for Azure DevOps pipeline events using Power Automate, eliminating manual project coordination.
- Architected an **automation framework using Playwright with TypeScript**, following **Page Object Model (POM)** and **CI/CD integration using Jenkins** for automated, parallel Playwright test execution.
- Leveraged **Playwright MCP** to enable AI-assisted browser automation and natural-language-driven test execution, improving productivity in web application testing.
- Proficient in using **Git** for source version control.
- Utilized **Postman** to validate API responses, ensuring accurate and reliable data exchanges.
- Implemented **Salesforce** solutions in both **Sales (CRM & CPQ)** and Service Cloud environments.

Sr. Quality Analyst

March 2018 to August 2021

Classic Informatics, Chandigarh

Projects: 360 Med Care, Fair Go Finance

- **REST API** testing using **Postman and Newman**, ensuring robust and reliable API functionality.
- Built **mock servers** in Postman to simulate API interactions for comprehensive testing.
- Managed version control using **Git/GitHub**, continuous integration with Jenkins, and project management and defect tracking with **JIRA (Zephyr)**.
- Conducted responsive testing using **BrowserStack, Lambda Test and Blisk**, ensuring optimal performance across various devices.
- Worked extensively on an **existing Selenium automation framework built using Java**, following the **Page Object Model (POM)** design pattern.

QA Engineer

October 2015 to March 2018

Wolters Kluwer, Chandigarh

Project: CCH iFirm

- Executed Regression, Integration, System, and Compatibility testing to ensure software quality.
- Led client requirement-gathering sessions; analysis improvements contributed to a **25% increase in project success rate**.

Software Tester

November 2011 to September 2015

Xequire Solutions, Chandigarh

Projects: Youth Enrollment System, Next Pawn, Lone Star Realty, Lewis Orthopedics

- Started a career as a manual test engineer, testing applications such as **YES, Next Pawn, Lone Star Realty**.
- Prepared **initial documents, SRS, summary reports** and **user manuals** to support both internal and external stakeholders.
- Worked within the **Waterfall Model**, adhering to its structured and sequential development process.

TECHNICAL LEADERSHIP & CONTRIBUTIONS

- **Technical Speaker:** Led multiple organization-wide knowledge-sharing sessions on automation strategies and modern testing tools.
- **Technical Blogger:** Actively publish technical content on automation testing, Playwright, and TypeScript to drive continuous learning within the QA community.
- **Mentorship:** Mentored 8+ junior engineers through formal mentorship programs.

CERTIFICATION

- ISTQB - Foundation Level (FL)
- Scrum Fundamentals Certified
- Generative AI Foundations (UpGrad)

AWARDS & RECOGNITION

- **HighQ Award - QualityAI**
 - Awarded for exceptional dedication to the pursuit of excellence and consistent high-quality performance.
- **QTribe Warrior – QualityAI**
 - Recognized for actively supporting and contributing to the “Automation QTribe” community.
- **First Prize – MVP Category, Code Games – Wolters Kluwer**
 - Won first prize in the Minimum Viable Product category for innovative solution design during the internal Code Games hackathon.

EDUCATION

- **Bachelor of Technology (B.Tech) - Information Technology**
Punjab Technical University (2007 – 2011) | **82%**
- **Senior Secondary (12th) – PSEB | 76%**
- **Secondary (10th) – PSEB | 87%**